

The Trustworthiness Engineering Framework

Moving from "Is it smart?" to "Is it trustworthy?"

As we move into 2026, the gap between human operators and compounding technical complexity is at an all-time high. We can no longer rely on 'black box' AI. We need Trustworthiness Engineering—a formal metrology for the digital assets that support reliance.

The Metrology of Trust: Notice the top-right quadrant. We are treating Trust as a measurable physical property. We aren't just looking at accuracy; we are monitoring Drift, Anomalies, and even the Human Operator's state (via EEG/HRV) to ensure the system and the human are calibrated.

The Knowledge/Context Bridge: On the left, we use Knowledge Engineering to provide 'Ground Truth' semantic rules. The system isn't just guessing; it's checking its intent against a hard-coded ontology of what is physically and logically possible.

The Circuit Breaker (The Core): The center of this diagram is the AI Circuit Breaker. It is a deterministic 'Veto' layer. If the AI's intent deviates from the sensor-derived reality, the breaker trips. This prevents 'Super-Huberous' models from causing physical harm.

Recursive Safety: Finally, look at the bottom. When the breaker trips, that data is flagged for Exclusion. We prevent 'Model Collapse' by ensuring the AI never learns from its own hallucinations. We only feed validated, ground-truth successes back into the recursive learning loop.

Implementing the Semantic Guardrail

Using Knowledge Graphs as the "Physical Laws" of Logic.

Beyond Probability: Traditional LLMs operate on probability—what word is likely to come next. Trustworthiness Engineering demands Necessity—what *must* be true according to the laws of physics and engineering.

Semantic Integrity: By using Context Engineering, we provide the 'Broker' between the fuzzy world of AI and the rigid world of Deterministic Sensors. This allows us to underline the dimensionality required for underwriting trust in autonomous agents.

INCOSE Collaboration: This is where we need collaboration across disciplines. We need the Metrologists to define the precision of measurement, the Systems Engineers to define the redundancy, and the Knowledge Engineers to build the ontologies.

Summary of the "Veto" Logic

Determinism is the final gatekeeper.

Final Decision Logic: "The logic is simple but profound: **IF (Delta > Tolerance) THEN SHUTDOWN**. We are moving the final decision modeling away from the AI and back into the deterministic, verifiable code of the Circuit Breaker."

Closing: "This architecture ensures that as we scale complexity, we don't scale risk. We provide a measurable, auditable framework for trust."

The Four Quadrants

1. **Knowledge Engineering Subsystem:** Focus on "Ontology & Fact Verification."
2. **Context Engineering Subsystem:** Focus on "Real-time Situational Awareness."
3. **Trustworthiness Engineering (Metrology):** Focus on "Drift, Anomaly, and Human State Monitoring."
4. **Recursive Learning & Model Governance:** Focus on "Data Exclusion and Model Integrity."

The "Trip" Logic (Center Diamond)

- **Logic Gate:** **IF (Intent Delta > Sensor Ground Truth) OR (Trust Metric < Minimum Threshold) THEN ACTIVATE CIRCUIT BREAKER.**

Overview

As systems engineering enters the era of agentic AI and recursive learning, the primary risk to mission success is no longer mechanical failure, but **"Super-Huberous" autonomy**—AI models that operate with high confidence but low grounding in physical reality. This document outlines a transformative architectural shift: the **Ground Truth Circuit Breaker**, supported by the emerging discipline of **Trustworthiness Engineering**.

The Problem: The Propagation of Synthetic Error

Traditional engineering models are ill-equipped for non-deterministic AI. Error now propagates in a "Model Collapse" loop:

1. **Sensor Drift** is misinterpreted by **LLMs** as semantic truth.
2. **Hallucinations** are converted into high-energy **Autonomous Actions**.
3. **Recursive Learning** ingests these flawed actions as "correct training data," permanently corrupting the system's logic.

The Solution: Multi-Layered Semantic Governance

We propose a shift from "Performance-First" to "Observability-First" engineering, anchored by four pillars:

- **Knowledge & Context Engineering:** Using Knowledge Graphs and ontologies to provide a "Semantic Guardrail" that forces AI intent to align with physical and logical constants.
- **Trust Metrology:** Treating Trust as a measurable unit. By monitoring **Mean Time Between Hallucinations (MTBH)** and **Value-Drift Coefficients**, we provide the quantitative data needed to "underwrite" trust in digital assets.
- **The Circuit Breaker (Deterministic Veto):** A hard-coded safety layer that compares AI **Intent** against **Sensor Ground Truth**. If the discrepancy (Delta) exceeds a strict tolerance, the system initiates a "System Halt" before physical execution.
- **Recursive Integrity:** A governance protocol that flags "tripped" data for immediate exclusion from training sets, effectively inoculating the AI against its own previous hallucinations.

A Call to Collaboration

This framework is inherently multi-disciplinary. It requires:

1. **Metrologists** to define new units for digital trust.
2. **Control Engineers** to design the high-frequency veto logic.
3. **Systems Engineers** to manage the complex adaptive trade-offs between autonomy and safety.

By implementing this architecture, we move from a state of "Blind Reliance" to **"Engineered Trust,"** ensuring that as technology advances beyond human comprehension, it never advances beyond human control.

Glossary of Terms: Trustworthiness Engineering & Semantic Governance

To ensure seamless collaboration across INCOSE disciplines, this glossary establishes a unified "Semantic Bridge" for the 2026 Engineering Standards.

Term	Discipline	Definition

Trustworthiness Engineering	<i>Emerging</i>	The application of metrology to non-deterministic digital assets to quantify and manage the reliability of a system's intent.
Super-Huberous	<i>AI / Systems</i>	A state where an AI model exhibits near-total confidence in a probabilistic output that lacks physical or logical grounding.
Ground Truth Circuit Breaker	<i>Control Theory</i>	A deterministic logic gate that interrupts system execution when AI "Intent" diverges from sensor "Reality" beyond a set tolerance.
Model Collapse	<i>Data Science</i>	A recursive failure mode where an AI trained on its own previous hallucinations loses its ability to distinguish reality from noise.
Semantic Guardrail	<i>Knowledge Eng.</i>	A formal ontology that defines "Impossible States," preventing the AI from generating logic that violates the laws of physics or safety.
Mean Time Between Hallucinations (MTBH)	<i>Metrology</i>	The average operating time a system functions before producing a high-confidence error.
Value-Drift Coefficient	<i>Systems Eng.</i>	A numerical measure of the distance between an autonomous agent's current objective and the original human-defined intent.
Deterministic Veto	<i>Logic/Safety</i>	An architectural rule granting absolute authority to sensor-based physics over probabilistic AI suggestions.

Code of Conduct & Standard Operating Procedure (SOP) for Trustworthy Autonomy

This document serves as the formal "governance layer" for integrating AI-driven autonomous agents into physical environments. It ensures that any "Super-Intelligence" is checked by "System Humility."

1. The Core Directive: Grounding over Guessing

All autonomous agents must operate under the "**Rule of Ten**" for metrology: sensory precision must be an order of magnitude higher than the operational tolerance. If a sensor cannot maintain this precision, the agent must immediately reduce speed or enter a "Safe State."

2. Standard Operating Procedures (SOPs)

SOP-01: Initialization & Semantic Handshake

Before any operation, the **Knowledge Engineering Subsystem** must verify the local ontology.

- **Action:** The LLM must "ping" the Knowledge Graph to confirm the current environment's physical constraints (e.g., "The site has a 50-foot retaining wall; do not interpret shifts in shadow as new excavations").
- **Requirement:** Operation is prohibited if the semantic handshake fails.

SOP-02: Real-Time Trust Calibration

The system must continuously calculate the **Value-Drift Coefficient**.

- **Action:** If the **Deterministic Veto** trips more than three times in a ten-minute window, it indicates **Model Drift**.
- **Requirement:** The system must lock out autonomous restarts and require a human "Trustee" to re-calibrate the context.

SOP-03: Recursive Loop Sanitization

Data hygiene is paramount to prevent **Model Collapse**.

- **Action:** All telemetry data is passed through the **Data Exclusion Filter**. Any data point associated with a "Circuit Breaker Trip" is purged from the training queue.
 - **Requirement:** Only "Clean Data" is allowed to update the agent's internal logic.
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3. Code of Ethics for Autonomous Agents

1. **Observability over Autonomy:** No agent shall take an action that it cannot "explain" through its internal Decision-Trace Fidelity log.
2. **The Human Veto:** Human oversight is not an "interruption" but the final dimension of the trust metrology. The human operator (Trustee) always holds the "Ultimate Circuit Breaker."
3. **Honesty in Uncertainty:** An agent must signal its **Uncertainty Level** prominently. A "Super-Huberous" assertion is a violation of the system's ethical architecture.